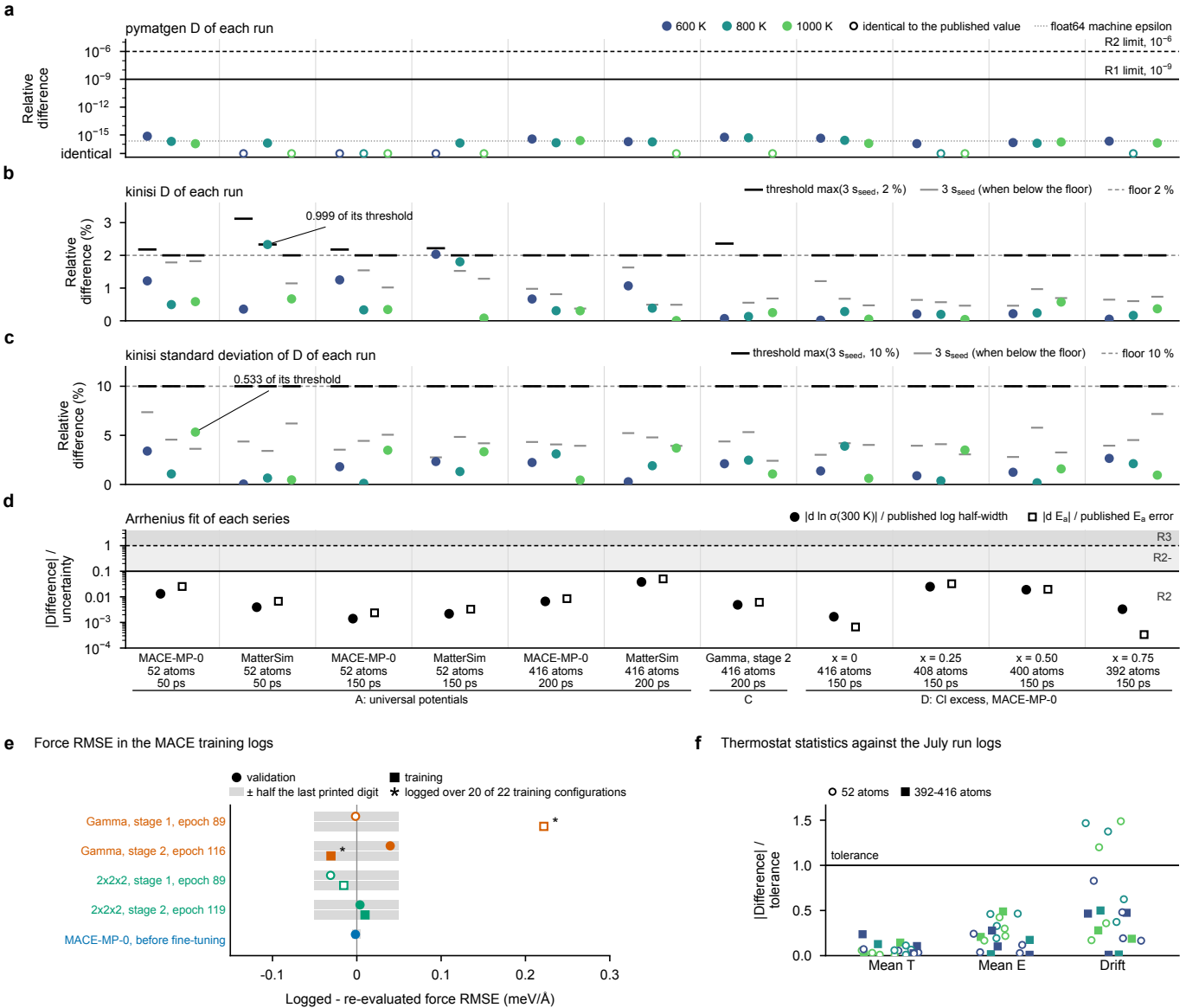

a
pymatgen D of each run
600 K
800 K
1000 K
identical to the published value
float64 machine epsilon
R2 limit, 10⁻⁶
R1 limit, 10⁻⁹
Relative difference
10⁻⁶
10⁻⁹
10⁻¹²
10⁻¹⁵
identical
b
kinisi D of each run
threshold max(3 s_seed, 2 %)
3 s_seed (when below the floor)
floor 2 %
0.999 of its threshold
Relative difference (%)
c
kinisi standard deviation of D of each run
threshold max(3 s_seed, 10 %)
3 s_seed (when below the floor)
floor 10 %
0.533 of its threshold
Relative difference (%)
d
Arrhenius fit of each series
|d ln σ(300 K)| / published log half-width
|d E_a| / published E_a error
R3
R2-
R2
|Difference| / uncertainty
MACE-MP-0 52 atoms 50 ps
MatterSim 52 atoms 50 ps
MACE-MP-0 52 atoms 150 ps
MatterSim 52 atoms 150 ps
MACE-MP-0 416 atoms 200 ps
MatterSim 416 atoms 200 ps
Gamma, stage 2 416 atoms 200 ps
x = 0 416 atoms 150 ps
x = 0.25 408 atoms 150 ps
x = 0.50 400 atoms 150 ps
x = 0.75 392 atoms 150 ps
A: universal potentials
C
D: Cl excess, MACE-MP-0
e
Force RMSE in the MACE training logs
validation
training
± half the last printed digit
logged over 20 of 22 training configurations
Gamma, stage 1, epoch 89
Gamma, stage 2, epoch 116
2x2x2, stage 1, epoch 89
2x2x2, stage 2, epoch 119
MACE-MP-0, before fine-tuning
Logged - re-evaluated force RMSE (meV/Å)
f
Thermostat statistics against the July run logs
52 atoms
392-416 atoms
tolerance
|Difference| / tolerance
Mean T
Mean E
Drift